

== What is Java

Java is • a programming language
• object Oriented
• Platform Independent

- Secure
- Robust

It was developed by James Gosling at sun Microsystems in 1995.

Today java belongs to Oracle.

Why Java Became popular?

Because of one sentence

"Write Once, Run Anywhere (WORA)"

meaning:

Write program once

Run it on windows / Linux / MacOS
without changing code.

How does Java program run?

Step 1 we wrote java program and save it as Main.java. This is called source code.
Because it is written by humans. Human can read it. Computer cannot.

Step 2 Compiler starts
java has compiler called javac
compiler reads Main.java and convert it into Main.class
This file contain Bytecode

What is Bytecode?

Bytecode is neither human language nor Machine language.

It is an intermediate language.

Java $\longrightarrow$ Bytecode $\longrightarrow$ Machine code

- Bytecode can run on every operating system
 $\Rightarrow$ That's why java is portable.

Why Doesn't compiler Directly Produce Machine code?

Program $\longrightarrow$ compiler $\longrightarrow$ Machine code (.exe)

This generated program mainly run only on windows.

Linux or MacOS cannot use that executable

- Java solved this problem

$\rightarrow$ Instead of generating machine code. Java generates Bytecode (Bytecode run anywhere).

What is Platform Independence?

Platform means OS eg. windows, linux, macos, android

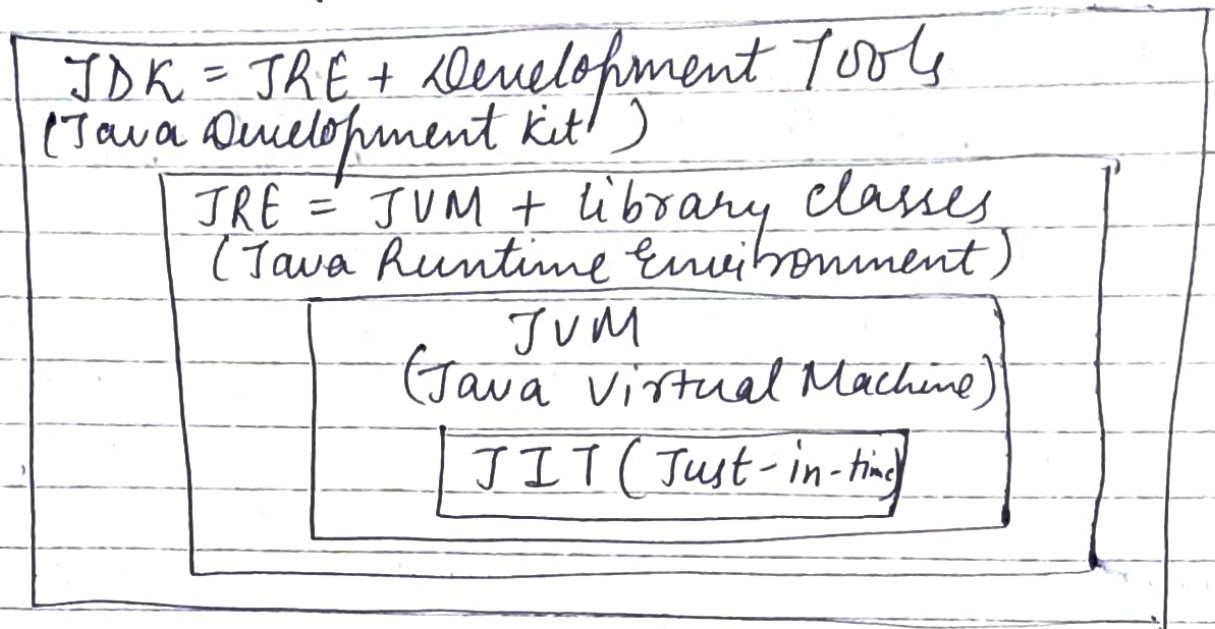
Suppose you wrote : Main.java
compiler creates Main.class

$\downarrow$
This file is exactly the same on every OS

- Only JVM changes

Hence, same class file
Different JVM
Different machine code
∴ Java is platform independent.

Architecture of Java

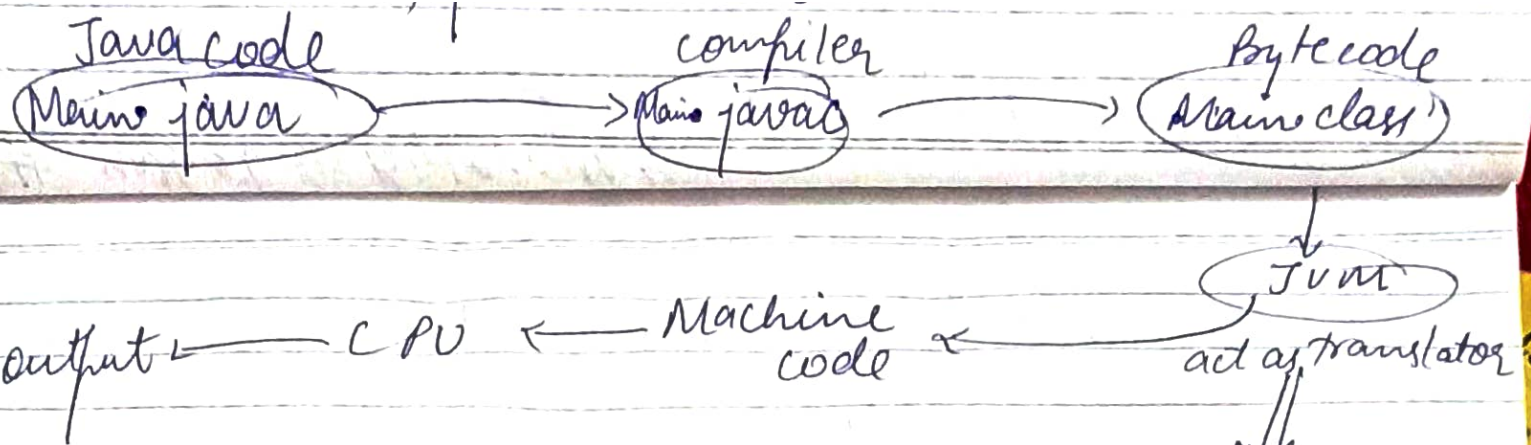


JVM (Java Virtual Machine) → no compiler
JVM is virtual computer.

It is responsible for

- Running Java Programs
- Converting Bytecode into Machine code
- Memory Management
- Garbage collection
- Security

Without JVM, java cannot run.



also checks whether
bytecode is safe before running
it.

Virus cannot directly attack
JVM easily.

JRE (Java Runtime Environment) → no compiler

~~many students confuse JRE & JRE~~

JRE = JVM + Libraries

contains thousands of ready-made
classes

eg: System.out.println();

comes from Java library

Same for: Scanner, String, Math, ArrayList.

Without JRE, java programs cannot run.

JDK (Java Development Kit) → contain compiler

JDK = JRE + development tools.

It is used by programmers.

It contains

- Compiler
- JRE
- JVM
- Debugger
- Javadoc
- Jar tool

Compile Time

computer checks

- syntax errors
- variable Declaration
- missing semicolon
- wrong method names

eg: `int a = "Hello";`

error (Program won't run)

Runtime

Program has already compiled
Now JVM executes it.

eg: `int a = 10;`
`int b = 0;`

`System.out.println(a/b);`

} compiler ✓

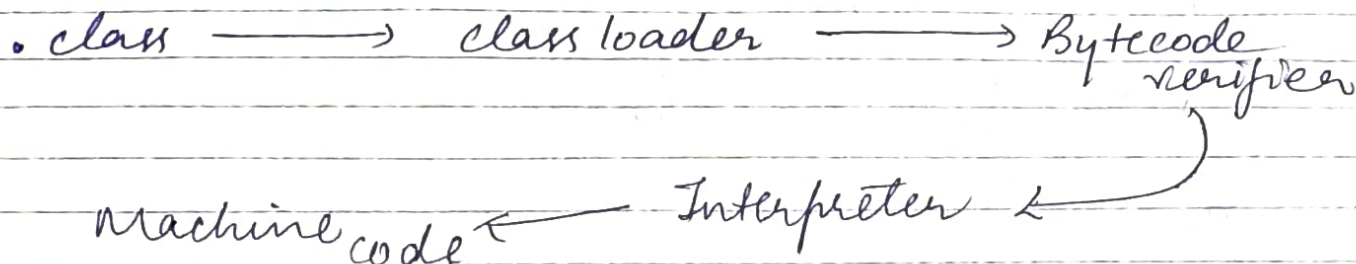
But while running

(ArithmeticException)

What happens After JVM gets Bytecode ?

JVM doesn't immediately run it

It first checks the file.



① Class loader = loads required classes.
loads `Main.class` into memory

It also loads, `String.class`
`Scanner.class`
`Object.class`
automatically.

② Bytecode verifier
checks Is this class safe?

- illegal code
- corrupted file
- invalid bytecode

} if unsafe program stops.

③ Interpreter
reads bytecode line by line (simple, but slow)

JIT (Just In Time compiler)

solution for interpreter

JIT compiles it into machine code.

JIT says "I'll translate it once."

Now, machine code is stored

next time run machine code directly
(Much faster)

